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Netflix Data Analysis and Visualization

This notebook solves the business case using the provided Netflix dataset.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
import datetime as dt
import warnings
warnings.filterwarnings('ignore')

# Load dataset
df = pd.read_csv('/content/sample_data/netflix_titles.csv')
df.head()
```

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],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n
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n   },\n   {\n   \"column\": \"release_year\",\n
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1969\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
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}\n   },\n   {\n   \"column\": \"duration\",\n
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min\",\n   \"177 min\"\n   ],\n   \"semantic_type\":
\"\",\n   \"description\": \"\"\n   }\n   },\n   {\n
\"column\": \"listed_in\",\n   \"properties\": {\n
\"dtype\": \"category\",\n   \"num_unique_values\": 514,\n
\"samples\": [\n   \"Crime TV Shows, International TV Shows, TV
Mysteries\",\n   \"Children & Family Movies, Classic Movies,
Dramas\"\n   ],\n   \"semantic_type\": \"\",\n
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small-town waitress makes the mistake of robbing a drug lord, putting
his life and newfound love in jeopardy.\",\n   \"Twelve-year-
old Calvin manages to join the navy and serves in the battle of
Guadalcanal. But when his age is revealed, the boy is sent to the
brig.\"\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   }\n]\
n}","type":"dataframe","variable_name":"df"}

1. Un-nesting the Columns (Comma-Separated Values)

```
def unnest_column(df, col):
    s = df.dropna(subset=[col])
    s = s.assign(**{col: s[col].str.split(', ')}).explode(col)
    return s
```

```python
df_cast = unnest_column(df, 'cast')
df_director = unnest_column(df, 'director')
df_genre = unnest_column(df, 'listed_in')

df_cast.head()
```

{"summary":"{\n \"name\": \"df_cast\",\n \"rows\": 64126,\n \"fields\": [\n   {\n   \"column\": \"show_id\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 7982,\n   \"samples\": [\n   \"s8467\",\n   \"s559\",\n   \"s5621\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"type\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 2,\n   \"samples\": [\n   \"Movie\",\n   \"TV Show\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"title\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 7982,\n   \"samples\": [\n   \"The Princess and the Frog\",\n   \"6 Bullets\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"director\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 4152,\n   \"samples\": [\n   \"Milton Horowitz\",\n   \"Brent Maddock\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"cast\",\n   \"properties\": {\n   \"dtype\": \"string\",\n   \"num_unique_values\": 36439,\n   \"samples\": [\n   \"Michael Biehn\",\n   \"Katie Jarvis\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"country\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 689,\n   \"samples\": [\n   \"Netherlands, Germany, Denmark, United Kingdom\",\n   \"United Kingdom, Canada, France, United States\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"date_added\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 1710,\n   \"samples\": [\n   \"August 10, 2017\",\n   \"May 19, 2019\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"release_year\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 8,\n   \"min\": 1942,\n   \"max\": 2021,\n   \"num_unique_values\": 72,\n   \"samples\": [\n   1996,\n   1977\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"rating\",\n   \"properties\": {\n   \"dtype\": \"category\",\n   \"num_unique_values\": 17,\n   \"samples\": [\n   \"TV-MA\",\n   \"PG\"\n

```
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
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\"Comedies, International Movies, Thrillers\"\n        ],\n
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\"num_unique_values\": 7956,\n        \"samples\": [\n          \"A
little girl must enter an orphanage after her parents are arrested,
but her life takes a dramatic turn when she escapes and meets a
model.\",\n          \"In a world where no one fibs, a screenwriter
gains fame and fortune \\u2013 and maybe the girl of his dreams \\
u2013 by saying things that aren't true.\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    }\n  ]\n}","type":"dataframe","variable_name":"df_cast"}
```


2. Handling Null Values

```
df.fillna({
    'director': 'Unknown Director',
    'cast': 'Unknown Cast',
    'country': 'Unknown Country',
    'rating': 'Unknown Rating',
    'date_added': 'Unknown Date'
}, inplace=True)
df.head()
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 8807,\n  \"fields\":
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\"s5495\"\n        ],\n        \"semantic_type\": \"\",\n
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Show\",\n          \"Movie\"\n        ],\n        \"semantic_type\":
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\"column\": \"title\",\n      \"properties\": {\n        \"dtype\":
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\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
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```

\"string\",\n   \"num_unique_values\": 4529,\n
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Colombari, Maurizio Lastrico, Alessandro Averone, Euridice Axen, Marco
Baliani, Pia Lanciotti, Giordano De Plano, Roberto Herlitzka, Tommaso
Ragno, Margherita Caviezel, Michele Morrone\",\n   \"Banky
Wellington, Rahama Sadau, Kanayo O. Kanayo, Ibrahim Suleiman, Michelle
Dede, Adesua Etomi, Hilda Dokubo, Akin Lewis\"\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
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8,\n   \"min\": 1925,\n   \"max\": 2021,\n
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}\n   },\n   {\n   \"column\": \"duration\",\n
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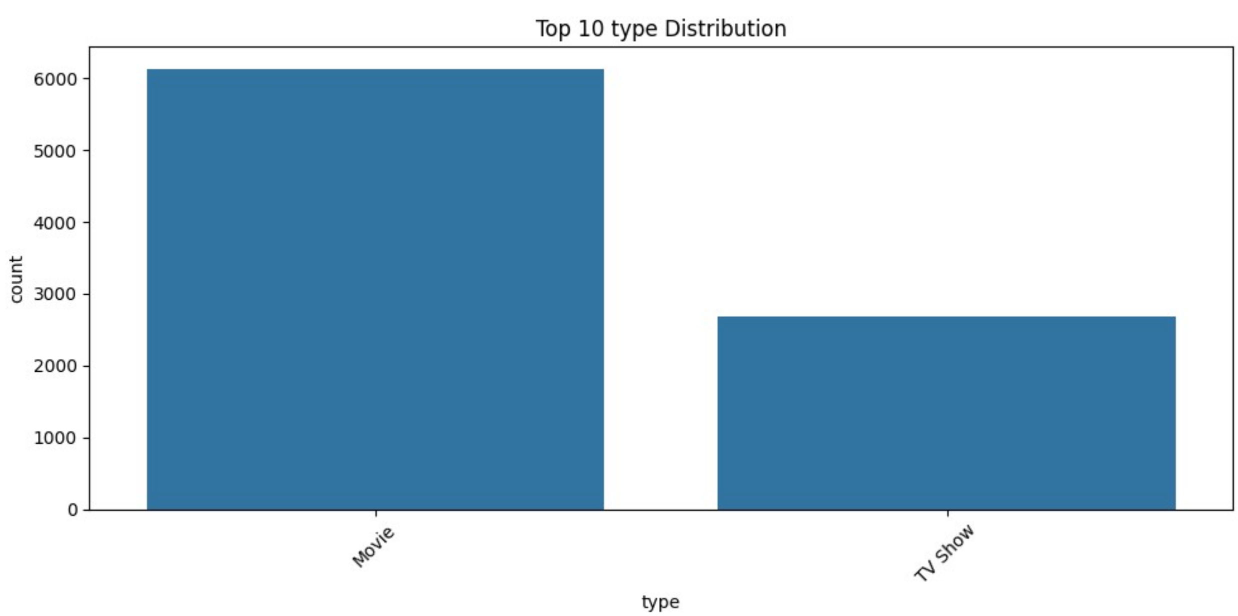
min\",\n   \"177 min\"\n   ],\n   \"semantic_type\":
\"\",\n   \"description\": \"\"\n   }\n   },\n   {\n
\"column\": \"listed_in\",\n   \"properties\": {\n
\"dtype\": \"category\",\n   \"num_unique_values\": 514,\n
\"samples\": [\n   \"Crime TV Shows, International TV Shows, TV
Mysteries\",\n   \"Children & Family Movies, Classic Movies,
Dramas\"\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
\"description\",\n   \"properties\": {\n   \"dtype\":
\"string\",\n   \"num_unique_values\": 8775,\n
\"samples\": [\n   \"A heedless teen drifter who falls for a
small-town waitress makes the mistake of robbing a drug lord, putting
his life and newfound love in jeopardy.\",\n   \"Twelve-year-

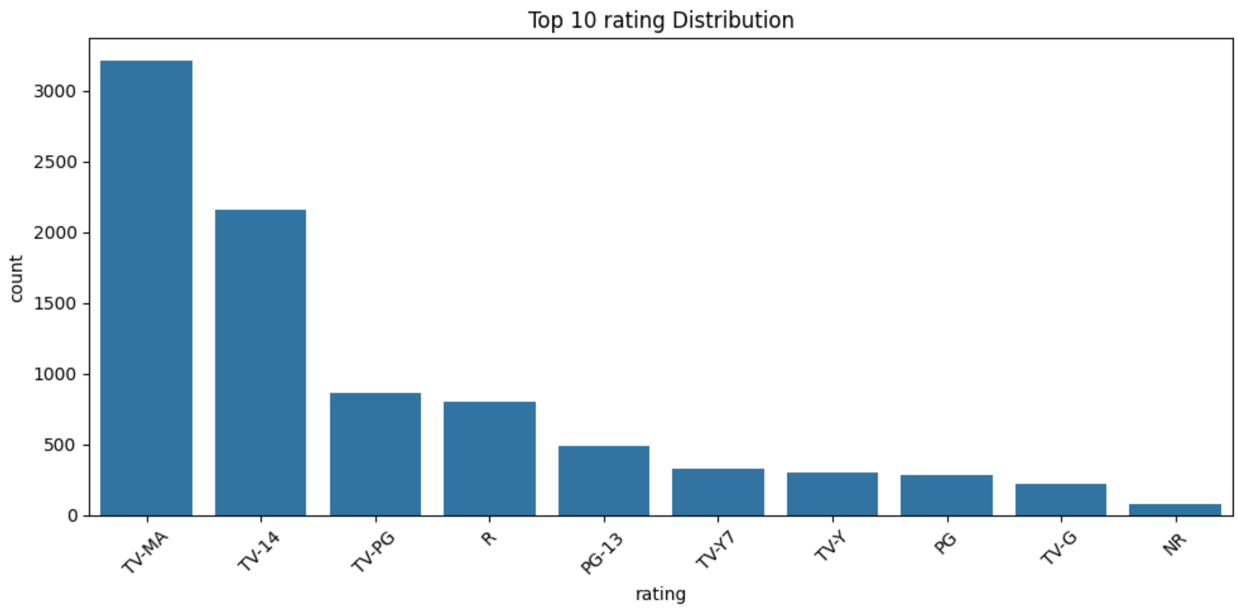
```
old Calvin manages to join the navy and serves in the battle of
Guadalcanal. But when his age is revealed, the boy is sent to the
brig.\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"df"}
```


3. Count of Each Categorical Variable (Graphical & Non-Graphical)

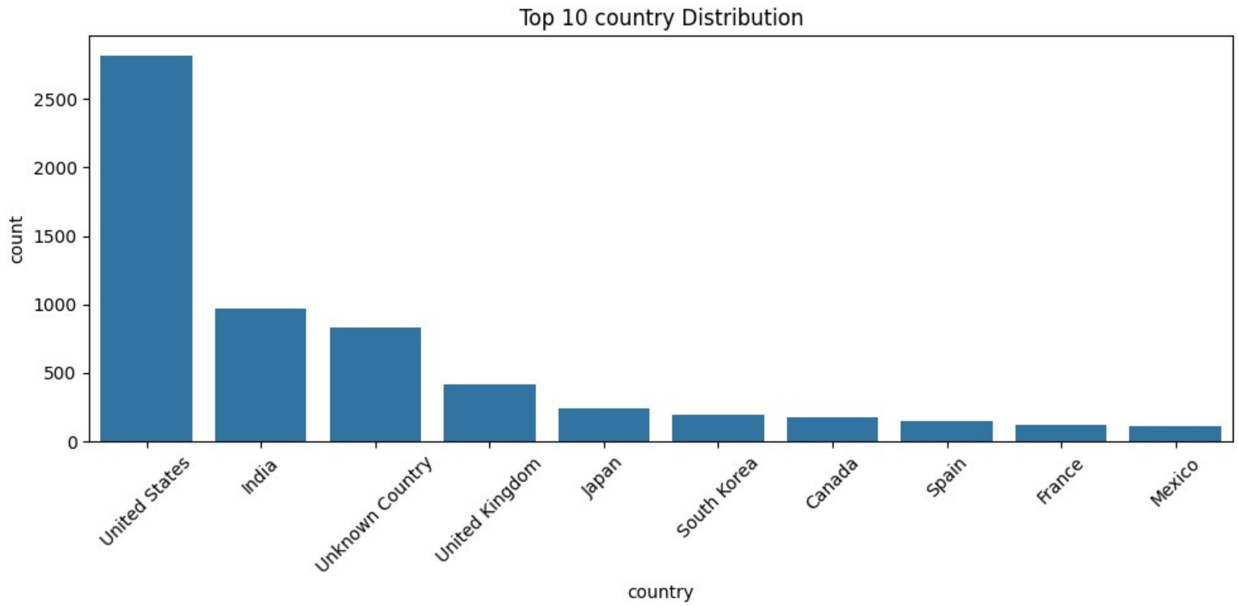
```
cat_columns = ['type', 'rating', 'country']
for col in cat_columns:
    print(f"\nValue Counts for {col}:")
    print(df[col].value_counts().head(10))

    plt.figure(figsize=(10,5))
    sns.countplot(data=df, x=col,
order=df[col].value_counts().index[:10])
    plt.xticks(rotation=45)
    plt.title(f'Top 10 {col} Distribution')
    plt.tight_layout()
    plt.show()


Value Counts for type:
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```



```
Value Counts for rating:
rating
TV-MA    3207
TV-14    2160
TV-PG     863
R         799
PG-13     490
TV-Y7     334
TV-Y      307
PG        287
TV-G      220
NR         80
Name: count, dtype: int64
```




```
Value Counts for country:
country
United States      2818
India               972
Unknown Country     831
United Kingdom      419
Japan               245
South Korea         199
Canada              181
Spain               145
France              124
Mexico              110
Name: count, dtype: int64
```

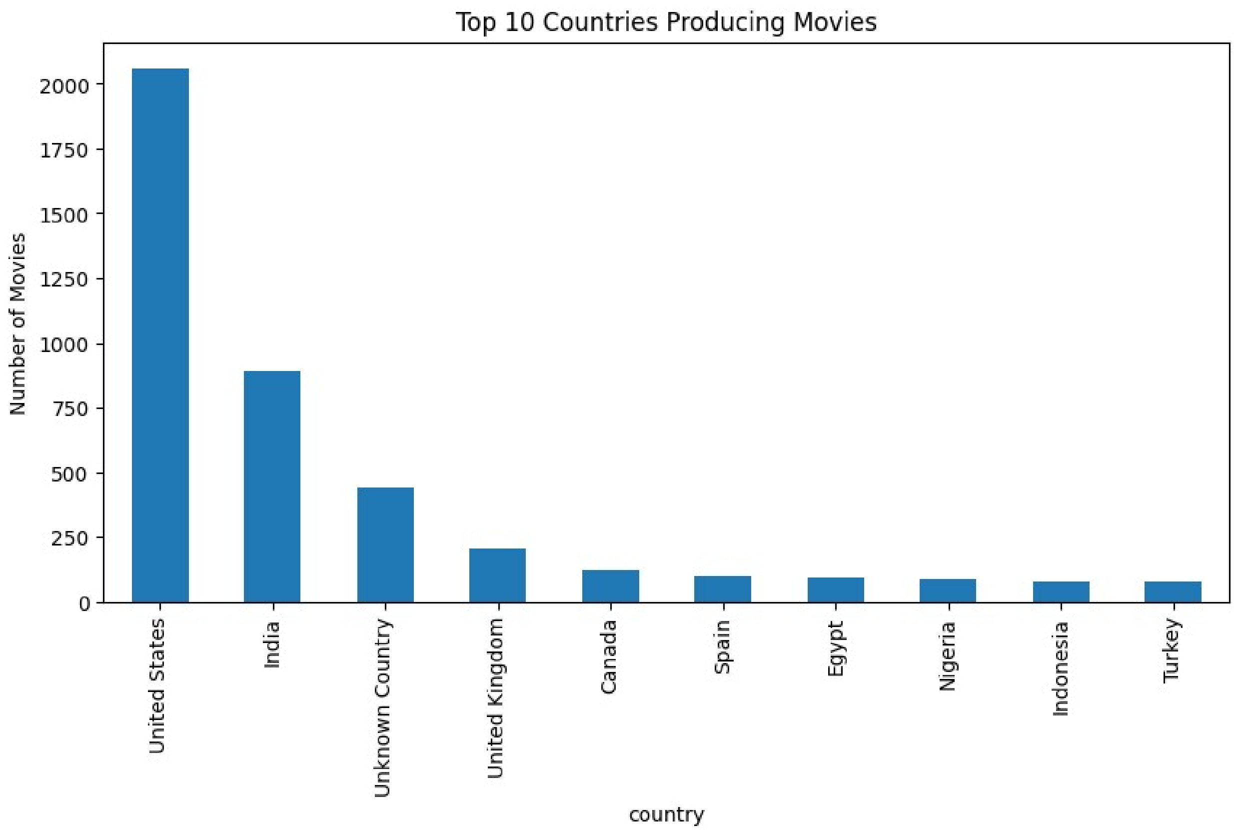


4. Comparison of TV Shows vs. Movies by Country

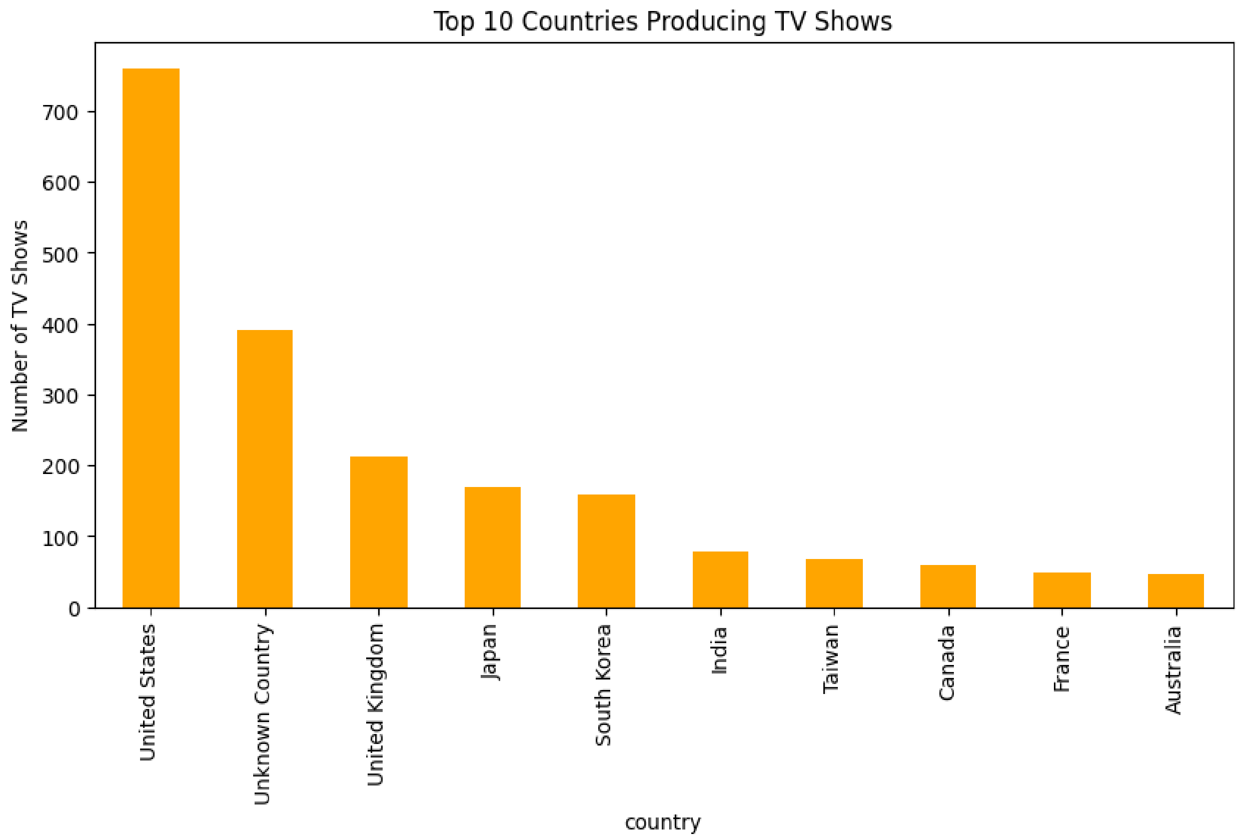
```
# Movies by country
top_movie_countries = df[df['type']=='Movie'].groupby('country')
['title'].count().sort_values(ascending=False).head(10)
top_tv_countries = df[df['type']=='TV Show'].groupby('country')
['title'].count().sort_values(ascending=False).head(10)

print("Top 10 Countries Producing Movies:")
print(top_movie_countries)

top_movie_countries.plot(kind='bar', title='Top 10 Countries Producing
Movies', figsize=(10,5))
plt.ylabel('Number of Movies')
plt.show()

print("\nTop 10 Countries Producing TV Shows:")
print(top_tv_countries)

top_tv_countries.plot(kind='bar', title='Top 10 Countries Producing TV
Shows', figsize=(10,5), color='orange')
plt.ylabel('Number of TV Shows')
plt.show()
```

```
Top 10 Countries Producing Movies:
country
United States      2058
India               893
Unknown Country     440
United Kingdom      206
Canada              122
```

```
Spain                    97
Egypt                    92
Nigeria                  86
Indonesia                77
Turkey                   76
Name: title, dtype: int64
```



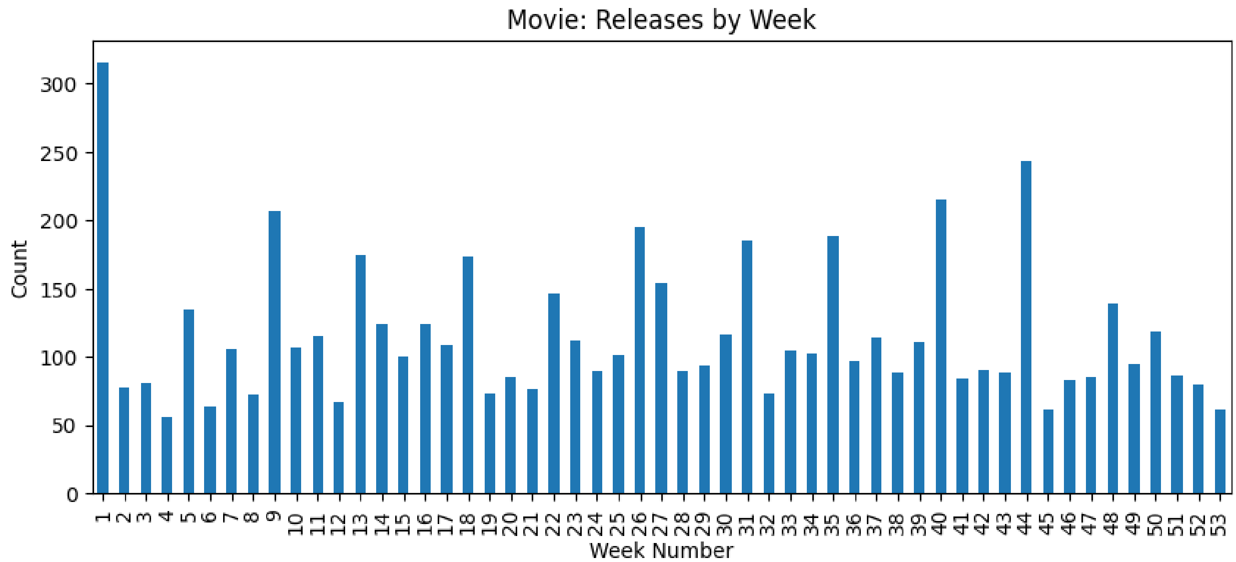

```
Top 10 Countries Producing TV Shows:
country
United States      760
Unknown Country    391
United Kingdom     213
Japan              169
South Korea        158
India               79
Taiwan              68
Canada              59
France              49
Australia           48
Name: title, dtype: int64
```

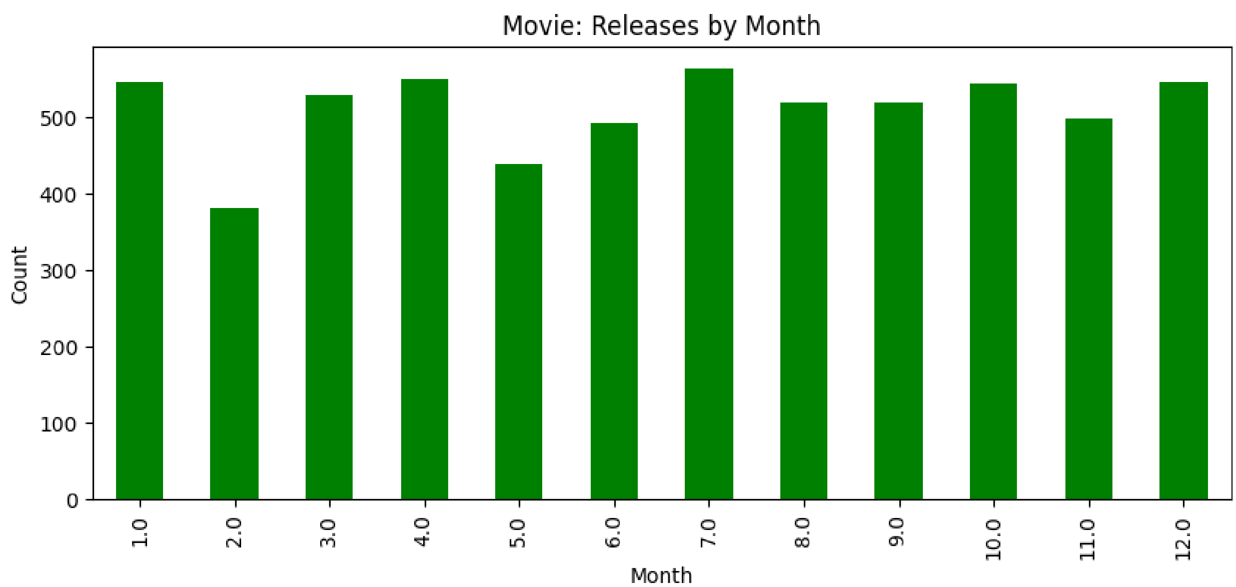


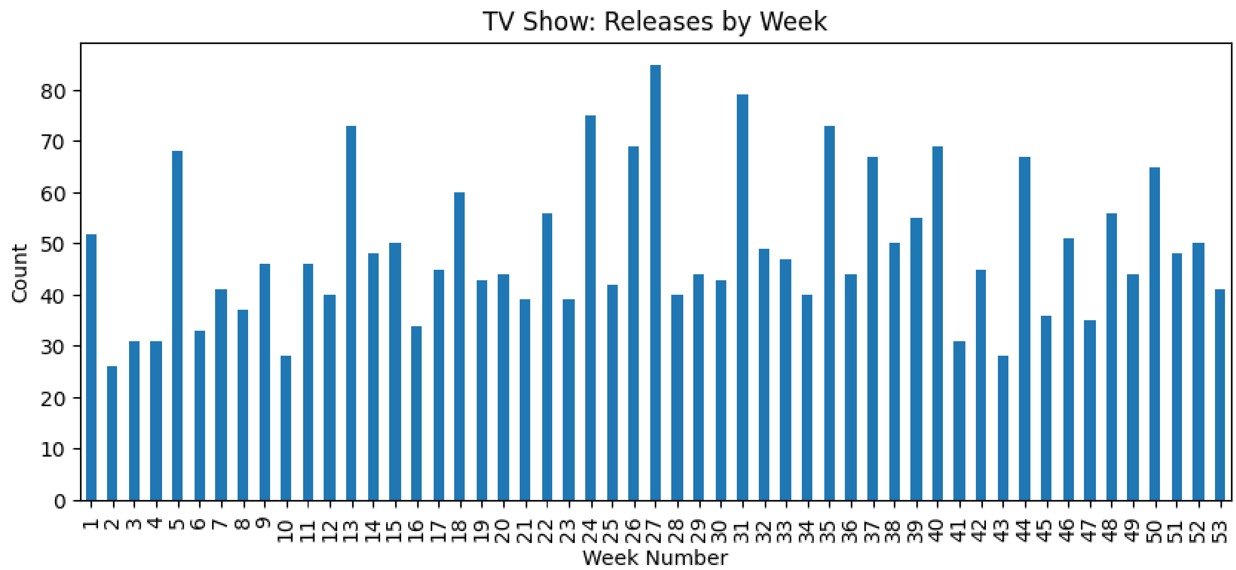
5. Best Time to Launch TV Shows or Movies

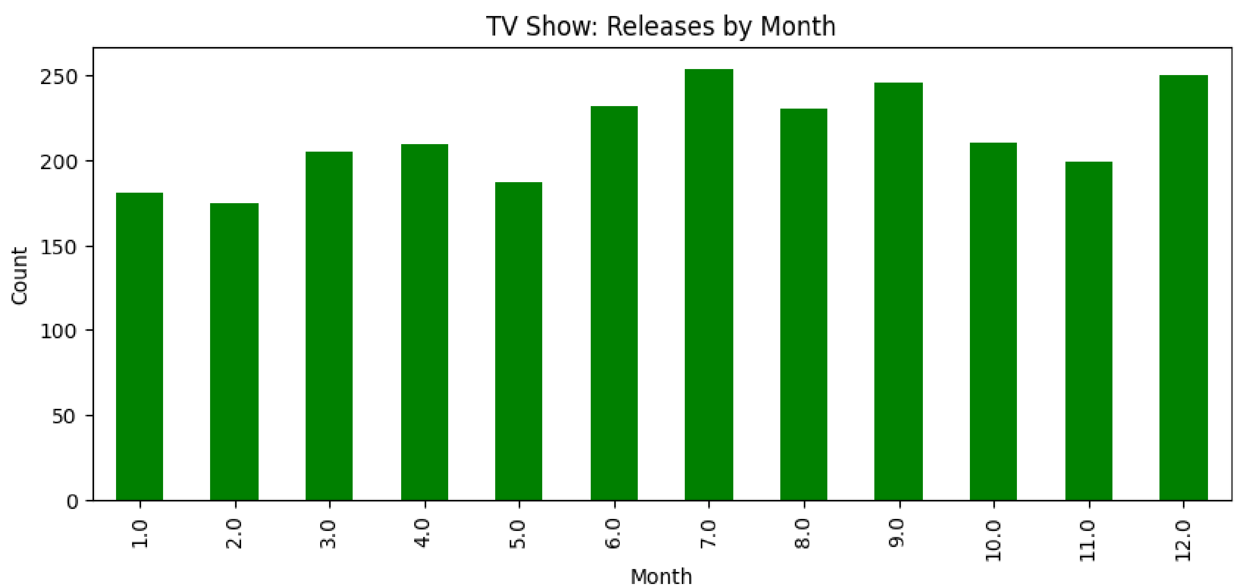
```
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
df['week'] = df['date_added'].dt.isocalendar().week
df['month'] = df['date_added'].dt.month

# Weekly and Monthly Launches
for t in ['Movie', 'TV Show']:
    df_temp = df[df['type']==t]
    week_counts = df_temp['week'].value_counts().sort_index()
    month_counts = df_temp['month'].value_counts().sort_index()

    plt.figure(figsize=(10,4))
    week_counts.plot(kind='bar')
    plt.title(f'{t}: Releases by Week')
    plt.xlabel('Week Number')
    plt.ylabel('Count')
    plt.show()

    plt.figure(figsize=(10,4))
    month_counts.plot(kind='bar', color='green')
    plt.title(f'{t}: Releases by Month')
    plt.xlabel('Month')
```

```
    plt.ylabel('Count')
    plt.show()
```







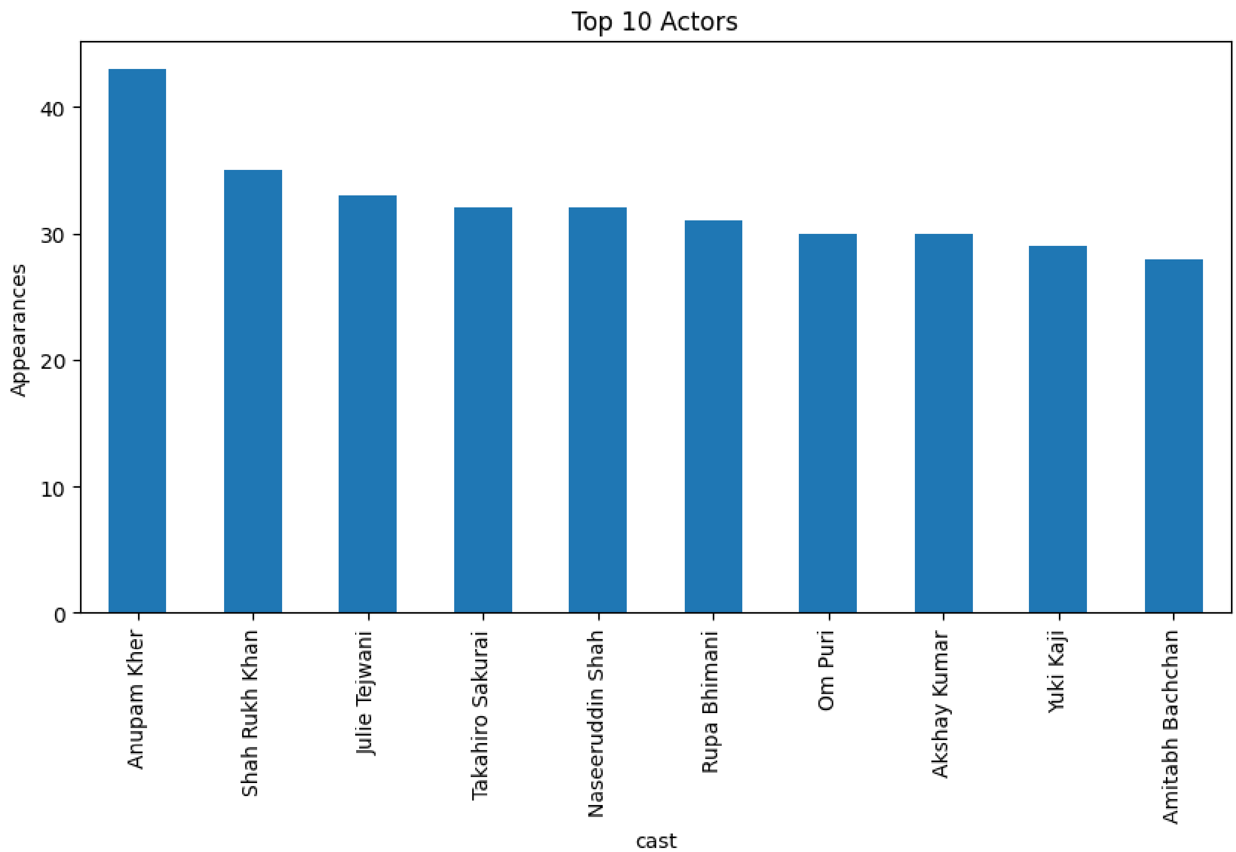


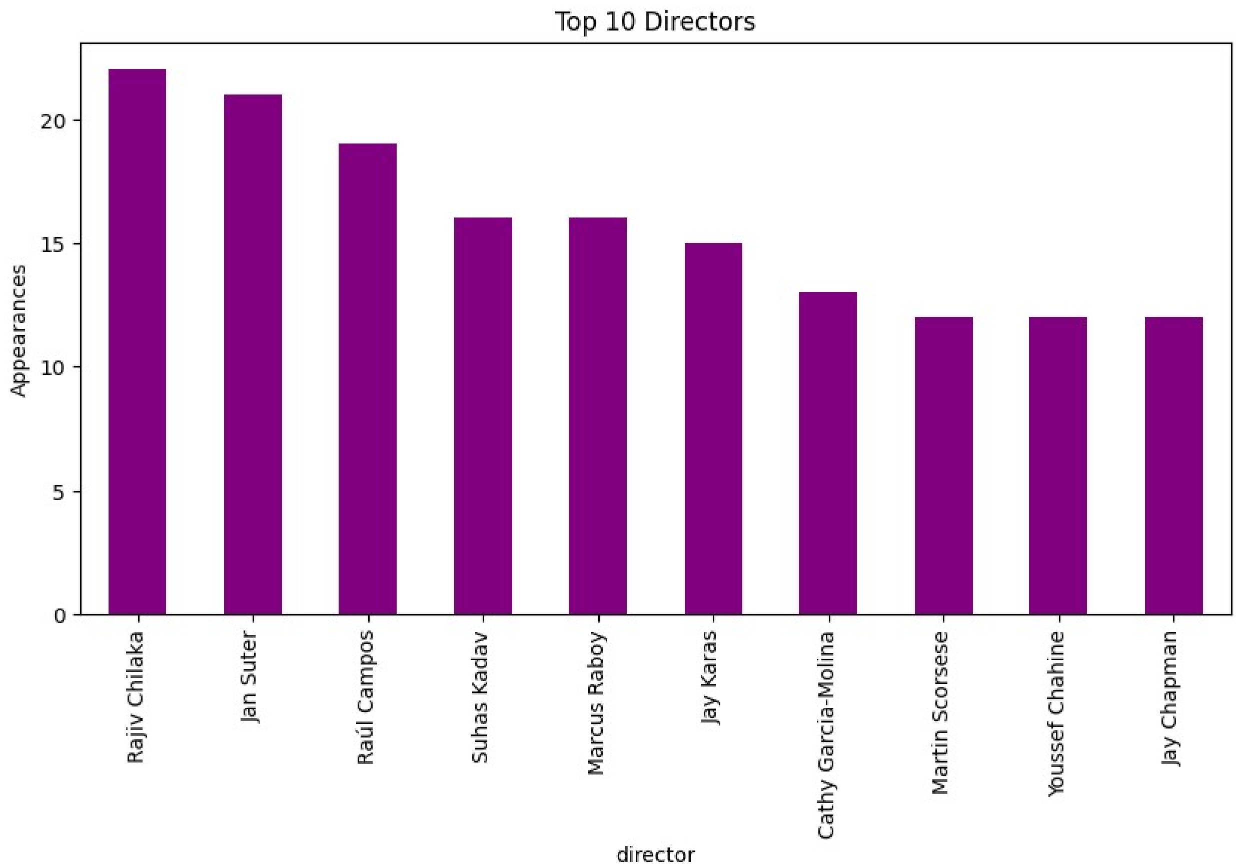
6. Top 10 Actors and Directors

```
top_actors = df_cast['cast'].value_counts().head(10)
top_directors = df_director['director'].value_counts().head(10)

print("Top 10 Actors:")
print(top_actors)

print("\nTop 10 Directors:")
print(top_directors)

top_actors.plot(kind='bar', figsize=(10,5), title='Top 10 Actors')
plt.ylabel('Appearances')
plt.show()
```

```
top_directors.plot(kind='bar', figsize=(10,5), color='purple',
title='Top 10 Directors')
plt.ylabel('Appearances')
plt.show()
```

```
Top 10 Actors:
cast
Anupam Kher          43
Shah Rukh Khan       35
Julie Tejwani        33
Takahiro Sakurai     32
Naseeruddin Shah     32
Rupa Bhimani         31
Om Puri              30
Akshay Kumar         30
Yuki Kaji            29
Amitabh Bachchan     28
Name: count, dtype: int64

Top 10 Directors:
director
Rajiv Chilaka             22
Jan Suter                 21
Raúl Campos               19
Suhas Kadav               16
Marcus Raboy              16
Jay Karas                 15
Cathy Garcia-Molina       13
Martin Scorsese           12
Youssef Chahine           12
Jay Chapman               12
Name: count, dtype: int64
```

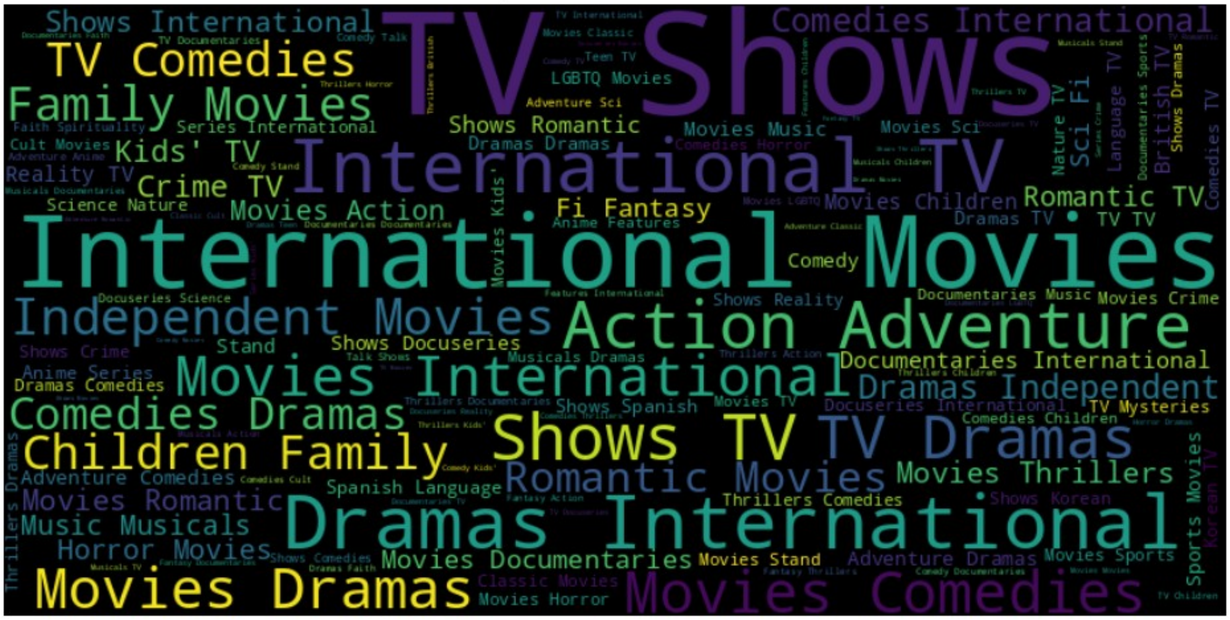

Top 10 Actors
Appearances
0
10
20
30
40
cast
Anupam Kher
Shah Rukh Khan
Julie Tejwani
Takahiro Sakurai
Naseeruddin Shah
Rupa Bhimani
Om Puri
Akshay Kumar
Yuki Kaji
Amitabh Bachchan



7. Genre Popularity Word Cloud

```
text = " ".join(df['listed_in'].dropna())
wordcloud = WordCloud(width=800, height=400,
background_color='black').generate(text)

plt.figure(figsize=(12,6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Popular Genres on Netflix')
plt.show()
```

Popular Genres on Netflix



8. Days Difference Between Release Year and Date Added

```python
df['year_added'] = df['date_added'].dt.year
df['diff'] = df['year_added'] - df['release_year']
df['diff'].dropna().astype(int).mode()
```

```
0    0
Name: diff, dtype: int64
```

Final Insights and Recommendations

- Most content is added between September and December.
- TV Shows and Movies both peak during the winter holiday season.
- The United States dominates in content production.
- The most popular genres include Drama, International Shows, and Comedies.
- Majority of content is added to Netflix 1 year after its official release.

These insights can help Netflix schedule new launches and guide production strategy.

THANK YOU !!!